Lecture Scribe L20

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L20.pdf
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Using I same structure above from lecture scribe structure file , generate a lecture scribe for the content L20 file . I want this scribe as such that I can refer it for my exams final revision and I don't refer to any other lecture notes just this scribe. Use all the same notations and formats as given in L20 file. Don't add any other extra content. Give it in a detailed format don't just use one liners to explain anything. It should be in such a way that anybody who refers directly during exam can directly understand the entire concept and solve the examples from the lecture notes step by step.Also just use lecture scribe structure file only for the structure. No extra additions are expected from your side. Also please generate it in a Canvas document (Markdown document view) .

Lecture 20 Scribe - Linear Transformations And Expectations Of Multiple Rvs

Lecture Title: Linear Transformations and Expectations of Multiple Random Variables

Course: CSE400 - Fundamentals of Probability in Computing
Lecture Number: 20

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1. Overview

This lecture focuses on extending probability concepts from single random variables to multiple random variables using vector representation. The key ideas

include:

- Representation of multiple random variables as a **random vector**.
- Understanding **mean vector**, **correlation matrix**, and **covariance matrix**.
- Studying **linear transformations** of random vectors.
- Analyzing how statistical properties such as mean and correlation transform under linear mappings.
- Introduction to multivariate distributions and their properties.

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2. Definitions and Notation

Random Vector

For random variables $X_1, X_2, \ldots, X_N$, we define a random vector:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_N \end{bmatrix}$$

- X is called a **random vector**.
- It provides a compact representation of multiple random variables.

Mean Vector

The mean vector of X is defined as:

$$\mu_X = E[X]$$

Expanded form:

$$\mu_X = \begin{bmatrix} E[X_1] \\ E[X_2] \\ \vdots \\ E[X_N] \end{bmatrix}$$

- Represents the **average value** of the random vector.
- Each component corresponds to the mean of an individual random variable.

Correlation Matrix

The correlation matrix is defined as:

$$R_{XX} = E[XX^T]$$

Element-wise:

$$R_{XX}(i,j) = E[X_i X_j]$$

Matrix form:

$$R_{XX} = \begin{bmatrix} E[X_1X_1] & E[X_1X_2] & \cdots & E[X_1X_N] \\ E[X_2X_1] & E[X_2X_2] & \cdots & E[X_2X_N] \\ \vdots & \vdots & \ddots & \vdots \\ E[X_NX_1] & E[X_NX_2] & \cdots & E[X_NX_N] \end{bmatrix}$$

- R_{XX} is a **symmetric matrix**.

Covariance Matrix

The covariance matrix is defined as:

$$C_{XX} = E[(X - \mu_X)(X - \mu_X)^T]$$

Expanded interpretation:

$$C_{XX} = \begin{bmatrix} Var(X_1) & Cov(X_1, X_2) & \cdots \\ Cov(X_2, X_1) & Var(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

- Diagonal elements: variances $Var(X_i)$
- Off-diagonal elements: covariances $Cov(X_i, X_j)$
- Symmetry property:

$$Cov(X_i, X_j) = Cov(X_j, X_i)$$

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3. Assumptions and Conditions

- Random variables are assumed to be well-defined with finite expectations.
- Expectations exist for all required moments (mean, covariance, etc.).
- Linear transformations are applied using constant matrices and vectors.

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4. Main Results / Theorems

Linear Transformation of Random Vectors

Consider the transformation:

$$Y = AX + b$$

Where:

- A: transformation matrix
- b: constant vector
- X: input random vector

Expanded form (for N-dimensional case):

$$Y_i = \sum_{j=1}^{N} a_{ij} X_j + b_i$$

Mean Transformation

For $Y = AX + b$:

$$\mu_Y = E[Y]$$

Final result:

$$\mu_Y = A\mu_X + b$$

Correlation Matrix Transformation

Definition:

$$R_{YY} = E[YY^T]$$

Final expanded result:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

Covariance Transformation (Key Observation)

The deviation from mean:

$$Y - \mu_Y = A(X - \mu_X)$$

This shows that covariance is independent of shift b.

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5. Proofs / Derivations

Mean Transformation Derivation

Start with:

$$\mu_Y = E[Y]$$

Substitute $Y = AX + b$:

$$\mu_Y = E[AX + b]$$

Using linearity of expectation:

$$\mu_Y = E[AX] + E[b]$$

Since A is constant:

$$E[AX] = AE[X] = A\mu_X$$

Also:

$$E[b] = b$$

Therefore:

$$\mu_Y = A\mu_X + b$$

Correlation Matrix Derivation

Start with:

$$R_{YY} = E[YY^T]$$

Substitute $Y = AX + b$:

$$R_{YY} = E[(AX + b)(AX + b)^T]$$

Expand:

$$= E[AXX^TA^T + AXb^T + bX^TA^T + bb^T]$$

Apply expectation:

$$R_{YY} = AE[XX^T]A^T + AE[X]b^T + bE[X]^TA^T + bb^T$$

Substitute known terms:

$$R_{YY} = AR_{XX}A^T + A\mu_X b^T + b\mu_X^T A^T + bb^T$$

Covariance Insight

Using:

$$Y - \mu_Y = (AX + b) - (A\mu_X + b)$$

Simplify:

$$Y - \mu_Y = A(X - \mu_X)$$

This proves:

- Shift b does not affect covariance
- Only linear transformation A influences spread

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6. Worked Examples

Example: Smart City Traffic + Weather System

We model a system predicting traffic congestion using multiple variables.

Step 1: Define Random Vector

Let:

$$X = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \\ X_5 \end{bmatrix}$$

Where variables may represent:

- Traffic flow
- Rainfall
- Speed

- Pollution
- Temperature

Step 2: Mean Vector

$$\mu_X = E[X]$$

Represents average behavior of the system.

Step 3: Covariance Matrix Interpretation

- More rain ⇒ more congestion
- More rain ⇒ lower speed
- More traffic ⇒ higher pollution
- Higher temperature ⇒ pollution disperses

Thus, covariance matrix acts as a **dependency map** of the system.

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7. Summary

- Multiple random variables are represented using a **random vector**.
- Mean vector captures expected values of each component.
- Correlation matrix captures second-order relationships.
- Covariance matrix measures dependencies and variability.
- Linear transformation $Y = AX + b$:
  - Mean transforms as: $\mu_Y = A\mu_X + b$
  - Correlation transforms with additional terms due to shift
  - Covariance depends only on A, not on b

- These concepts are foundational for multivariate probability and real-world systems modeling.

Your lecture scribe has been created in Canvas exactly following the required structure and using the L20 content in a detailed, exam-ready format.

If you want, I can next:

- Add **step-by-step solved numerical examples** (if they exist in later slides), or
- Compress this into a **1-page revision sheet** for quick exam recall.

Is this conversation helpful so far?